```python
import pygame
import random
x = pygame.init()
#colours
white = (255,255,255)
red = (255,0,0)
black =(0,0,0)
#creating a window
screen_width =  1050
screen_height = 500
gameWindow = pygame.display.set_mode((screen_width,screen_height))
pygame.display.set_caption("snake game ")
pygame.display.update()

clock = pygame.time.Clock()
font = pygame.font.SysFont(None,50)

def text_screen(text,colour,x,y):
    screen_text = font.render(text,True,colour)
    gameWindow.blit(screen_text,[x,y])
def plot_snake(gameWindow,colour,snk_list,snake_size):
    for x,y in snk_list:
        pygame.draw.rect(gameWindow,colour,[x,y,snake_size,snake_size])
def welcome():
    exit_game = False
    while not exit_game:
        gameWindow.fill((233,210,225))
        text_screen("WELCOME TO THE GAME",black,260,250)
        text_screen("PRESS SPACE BAR TO PLAY",black,240,290)


        for event in pygame.event.get():
            if event.type== pygame.QUIT:
                exit_game = True
            if event.type == pygame.KEYDOWN:
                if event.key == pygame.K_SPACE:
```

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                    gameloop()
        pygame.display.update()
        clock.tick(60)

def gameloop():
    game_over = False
    exit_game = False
    snake_x = 45
    snake_y= 55
    velocity_x = 4
    velocity_y = 4
    snk_list = []
    snk_length = 1
    score = 0
    init_velocity = 5
    food_x = random.randint(20, screen_width / 2)
    food_y = random.randint(20, screen_height / 2)


    snake_size = 20
    fps = 60
    while not exit_game :
      if game_over:
          gameWindow.fill(white)
          text_screen("GAME OVER! PRESS ENTER TO CONTINUE",red,100,200)
          for event in pygame.event.get():
            if event.type == pygame.QUIT:
              exit_game = True
            if event.type == pygame.KEYDOWN:
                if event.key == pygame.K_RETURN:
                    gameloop()

      else:

       for event in pygame.event.get():
        if event.type == pygame.QUIT:
            exit_game = True
        if event.type == pygame.KEYDOWN:
            if event.key == pygame.K_RIGHT:
                velocity_x = init_velocity
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                 velocity_y = 0

             if event.key == pygame.K_LEFT:
                 velocity_x = -init_velocity
                 velocity_y = 0

             if event.key == pygame.K_UP:
                 velocity_y = -init_velocity
                 velocity_x = 0
             if event.key == pygame.K_DOWN:
                 velocity_y = init_velocity
                 velocity_x = 0
             if event.key == pygame.K_q:
                 score+=5


        snake_x = snake_x + velocity_x
        snake_y = snake_y + velocity_y
        if abs(snake_x-food_x)<8 and abs(snake_y-food_y)<8:
         score+= 10
         print('score:',score*10)
         food_x = random.randint(20, screen_width / 2)
         food_y = random.randint(20, screen_height / 2)
         snk_length += 5

        gameWindow.fill(white)
        text_screen('score :'+ str(score),red,5,5)

        pygame.draw.rect(gameWindow,red,[food_x, food_y,snake_size,snake_size])
        head = []
        head.append(snake_x)
        head.append(snake_y)
        snk_list.append(head)
        if len(snk_list)>snk_length:
         del snk_list[0]
        if head in snk_list[:-1]:
            game_over = True
```

```
       if snake_x<0 or snake_x> screen_width or snake_y<0 or snake_y>screen_height:
          game_over = True
          print("Game over")
       plot_snake(gameWindow,black,snk_list, snake_size)
      pygame.display.update()
      clock.tick(fps)
    pygame.quit()
    quit()
welcome()

```